

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

23AID302 – Big Data Analytics

Course Project – Basic Requirements

AY : 2026-27

1. Project Title:

Should reflect analytics-based solutions on large datasets.

E.g :

- Big Data Analytics for Predictive Maintenance in IoT Systems
- Sentiment Analysis of Twitter Data Using Spark

2. Problem Statement

Must involve the following:

- **Volume:** Very large datasets (GBs, TBs)
- **Velocity:** Streaming data or real-time processing
- **Variety:** Structured, semi-structured, or unstructured data (any two to be included)
- **Veracity:** Handling noisy/incomplete/inconsistent data

3. Technical Requirements

- Hadoop (HDFS, MapReduce)
- Apache Spark (RDD/DataFrame/Dataset API)
- Scala & Python
- Spark ML

Datasets :

Must use **large open datasets** from sources such as:

- Kaggle
- Twitter API ., etc.,

4. System Features

- Data acquisition and pre-processing (cleaning, transformation)
- Batch or streaming data processing
- Application of analytical models (e.g., clustering, classification)
- Performance tuning (e.g., partitioning, caching in Spark)
- Visualization and dashboarding

SAMPLE PROJECT TITLES

- Real-Time Sentiment Analysis on Twitter Data using Spark Streaming
- Social Media Fake News Detection using Big Data Techniques
- Cancer Prediction using Genomic Big Data and Spark MLlib
- Credit Card Fraud Detection using Big Data and Machine Learning
- Energy Consumption Forecasting in Smart Homes Using Big Data.
- Network Intrusion Detection using Big Data Analytics
- Student Performance Prediction using Big Data Techniques

Project Domains - Samples

Domain	Example Project
Healthcare	Disease Prediction
Finance	Credit Risk Analysis
Cyber Security	Network Intrusion Detection
Agriculture	Crop Yield Prediction
Retail	Customer Purchase Analysis
Education	Student Performance Prediction
Transportation	Traffic Accident Analysis
E-commerce	Product Recommendation
Banking	Loan Approval Prediction
Social Media	Sentiment Analysis

Architecture :

